

Sampling: from unit hypercube to any pdf

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Outline

Introduction

The uniform distribution $\mathcal{U}(0, 1)$

From $\mathbf{u} \sim \mathcal{U}(0, 1)^d$ to $\mathbf{x} \sim p_{x_1} \times \cdots \times p_{x_d}$

From $\mathbf{u} \sim \mathcal{U}(0, 1)^d$ to $\mathbf{x} \sim p_{\mathbf{x}} \neq p_{x_1} \times \cdots \times p_{x_d}$

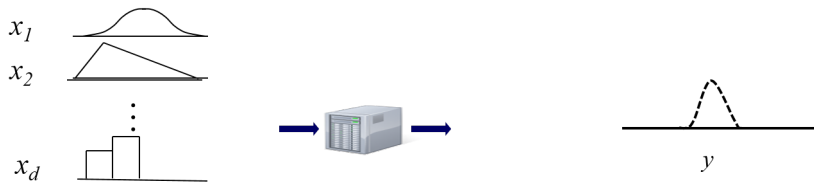
Cholesky transformation

Rosenblatt transformation

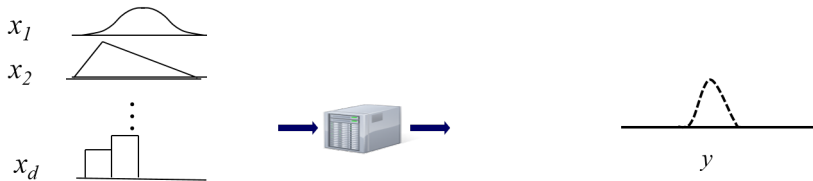
Conclusion

Uncertainty Analysis by Monte Carlo simulations

Let $y = \mathcal{M}(\mathbf{x})$ be the scalar model response of interest, with $\mathbf{x} = (x_1, x_2, \dots, x_d) \sim p_{\mathbf{x}}$.



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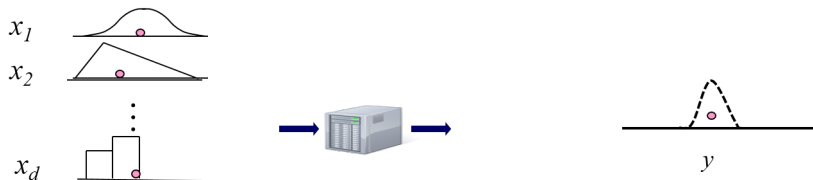
Goal: Estimate p_y , the probability density function (pdf) of the model response of interest, knowing $p_{\mathbf{x}}$

Mean: Monte Carlo simulations

Monte Carlo simulations

Monte Carlo simulations are a numerical way to propagate the input uncertainty into the model.

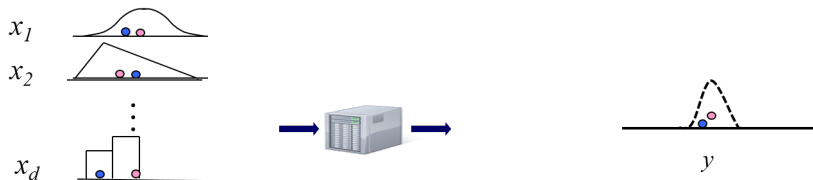
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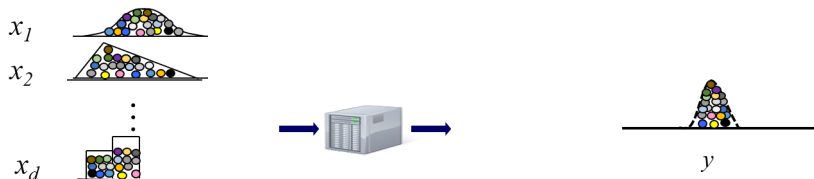
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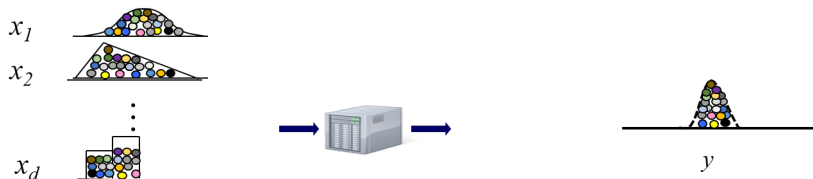
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Then, p_y and other statistics of y can be inferred from the Monte Carlo sample.

How to sample $\mathbf{x} \sim p_{\mathbf{x}}$?

Transform $\mathbf{u} \sim \mathcal{U}(0, 1)^d$ into $\mathbf{x} \sim p_{\mathbf{x}}$

$$u \sim \mathcal{U}(0, 1)$$

The uniform distribution
 $\mathcal{U}(0, 1)$

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Definition: A random variable (RV) u uniformly distributed over $(0, 1)$ has a probability density function (pdf) defined as follows,

$$p_u(u) = \begin{cases} 1 & \text{if } u \in (0, 1) \\ 0 & \text{otherwise} \end{cases}$$

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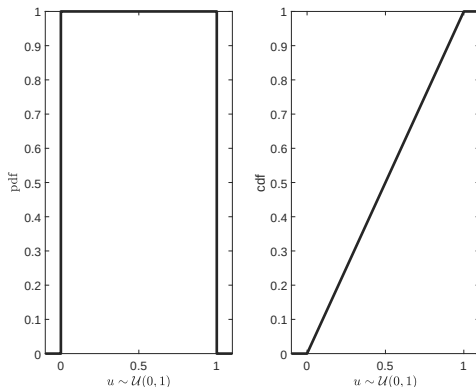
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Its cumulative density function (cdf) is,

$$F_u(u) = \int_{-\infty}^u p_x dx = \begin{cases} 0 & \text{if } u < 0 \\ u & \text{if } u \in (0, 1) \\ 1 & \text{if } u > 1 \end{cases}$$

u is completely defined by p_u or F_u .

The uniform distribution $\mathcal{U}(0, 1)$



Most of the programming languages contain by default a pseudo-random generator for $\mathcal{U}(0, 1)$. For example,

Matlab/Octave — `> rand`

R — `> runif`

$$\mathbf{u} \sim \mathcal{U}(0, 1)^d \rightarrow \mathbf{x} \sim p_{x_1} \times \cdots \times p_{x_d}$$

From $\mathbf{u} \sim \mathcal{U}(0, 1)^d$
to
 $\mathbf{x} \sim p_{x_1} \times \cdots \times p_{x_d}$

The Integral Transform

The integral transform: Let $\mathbf{x} \sim p_{x_1} \times \cdots \times p_{x_d}$ be a random vector of **independent** RVs arbitrary distributed, and $F_{x_1}, \dots, F_{x_d}$ be their associated cdfs.

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From a random vector $\mathbf{u} \sim \mathcal{U}(0, 1)^d$, it is straightforward to derive $\mathbf{x}$ with the following integral transformation,

$$x_i = F_{x_i}^{-1}(u_i) \quad (1)$$

with $i = 1, \dots, d$.

Given a sample $\mathbf{U}$ of $\mathbf{u}$ Eq.(1) provides a sample $\mathbf{X}$ of $\mathbf{x}$.

The Integral Transform

Some analytical integral transforms,

- ▶ if $x \sim \mathcal{U}(a, b)$ then $x = u(b - a) + a$
- ▶ if $x \sim \mathcal{N}(\mu, \sigma^2) = (2\pi\sigma^2)^{-1/2}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, then
 $x = \sigma\sqrt{2}\text{erf}^{-1}(2u - 1) + \mu$, erf is the error function.
- ▶ if $x \sim \mathcal{T}(a, b, c)$, $a < c < b$, then
$$x = \begin{cases} a + \sqrt{u(b-a)(c-a)} & \text{if } 0 < u < \frac{c-a}{b-a} \\ b - \sqrt{(1-u)(b-a)(b-c)} & \text{otherwise} \end{cases}.$$

Useful functions:

Matlab $\rightarrow$ erfinv ($= \text{erf}^{-1}$), gaminv (inverse cdf of Γ law),

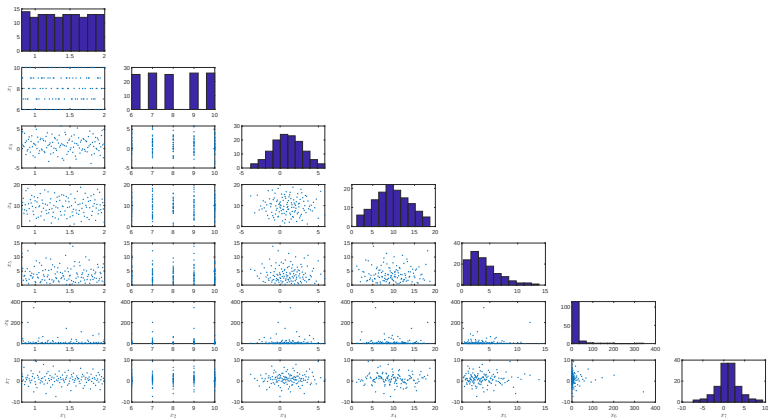
R $\rightarrow$ qnorm ($= \sqrt{2}\text{erf}^{-1}(2u - 1)$), qgamma (inverse cdf of Γ law),

The Integral Transform

Exercise

Exercise 1: Set $N = 128$, $d = 7$, and generate $\mathbf{U}$ a random sample of $\mathcal{U}(0, 1)^7$. From the sample $\mathbf{U}$, deduce the sample $\mathbf{X}$ such that

1. $\mathcal{U}(x_1|0.8, 2)$
2. $\mathcal{DU}(x_2|6, 10)$
3. $\mathcal{N}(x_3|1, 2^2)$
4. $\mathcal{T}(x_4|0, 20, 9)$
5. $\mathcal{N}(\ln(x_5)|1, 2^2)$, i.e. Log-Normal distribution



Uncertainty analysis of the borehole model

taken from <https://www.sfu.ca/~ssurjano/>

Practicum

The function models water flow through a borehole. The response of interest φ is the flow rate (in m^3/yr) defined as follows,

$$\varphi(\mathbf{x}) = \frac{2\pi T_u (H_u - H_l)}{\ln(r/r_w) \left(1 + \frac{2LT_u}{\ln(r/r_w)r_w^2 K_w} + \frac{T_u}{T_l} \right)}$$

The uncertain input variables, assumed **independent** of each other, are $\mathbf{x} = (r_w, r, T_u, H_u, T_l, H_l, L, K_w)$. Their marginal pdfs are given below,

r_w	Radius of Borehole (m)	$\mathcal{N}(r_w 0.11, 0.005^2)$
r	Radius of Influence (m)	$\mathcal{N}(\ln(r) 7.71, 1)$
T_u	Transmittivity of Upper Aquifer (m^2/yr)	$\mathcal{T}(T_u 63\,070, 115\,600, 100\,000)$
H_u	Pressure Head of Upper Aquifer (m)	$\mathcal{N}(\ln(H_u) 6.95, 0.0167^2)$
T_l	Transmittivity of Lower Aquifer (m^2/yr)	$\mathcal{U}(T_l 63, 116)$
H_l	Pressure Head of Lower Aquifer (m)	$\mathcal{N}(\ln(H_l) 6.6, 0.033^2)$
L	Length of Borehole (m)	$\mathcal{U}(L 1\,120, 1\,680)$
K_w	Hydraulic Conductivity (m/yr)	$\mathcal{U}(K_w 9\,855, 12\,045)$

Propagate the input uncertainty into the model response with $N = 256$. Estimate the mathematical expectation and variance of the model response.

$$\mathbf{u} \sim \mathcal{U}(0, 1)^d \rightarrow \mathcal{N}(\xi | \mathbf{0}, \mathbf{I}_d) \rightarrow \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \mathbf{C})$$

From $\mathbf{u} \sim \mathcal{U}(0, 1)^d$
to
 $\mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \mathbf{C})$

Cholesky Transform

Step 1: By using the integral transform it is straightforward to transform the u -variables into $\mathcal{N}(\boldsymbol{\xi}|\mathbf{0}, \mathbf{I}_d)$ as the ξ -variables are independent on each other: $x_i = \sqrt{2}\text{erf}(2u_i - 1)$

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Step 2: By denoting $\mathbf{U}_p$ the upper Cholesky matrix of the covariance matrix $\mathbf{C}$, that is:

$$\mathbf{C} = \mathbf{U}_p^t \mathbf{U}_p$$

the x -variables ($\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C})$) are obtained as follows:

$$\mathbf{x} = \boldsymbol{\mu} + \boldsymbol{\xi} \mathbf{U}_p$$

Cholesky Transform

Exercises

Exercise 2: Generate normally distributed samples

Case 1

$$\boldsymbol{\mu} = (1, 2) \text{ and } \mathbf{C} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{such that } \mathbf{U}_p = \begin{bmatrix} \sqrt{2} & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$$

N.B.: The Cholesky transform is not unique. The following case is the same problem but provides different samples

$$\boldsymbol{\mu} = (2, 1) \text{ and } \mathbf{C} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \quad \text{such that } \mathbf{U}_p = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Case 2

$$\boldsymbol{\mu} = (0, 0, 0) \text{ and } \mathbf{C} = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix}, \quad \mathbf{U}_p = \begin{bmatrix} \sqrt{3} & \frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{3} \\ 0 & \sqrt{\frac{5}{3}} & \frac{\sqrt{15}}{15} \\ 0 & 0 & \sqrt{\frac{3}{5}} \end{bmatrix}$$

$$\mathbf{u} \sim \mathcal{U}(0, 1)^d \rightarrow \mathbf{x} \sim p_{\mathbf{x}} \neq p_{x_1} \times \cdots \times p_{x_d}$$

From $\mathbf{u} \sim \mathcal{U}(0, 1)^d$
to

$$p_{\mathbf{x}} = p_{x_{i_1}} p_{x_{i_2}|x_{i_1}} \cdots p_{x_{i_d}|x_{i_1} \cdots x_{i_{d-1}}}$$

Rosenblatt Transform

The Rosenblatt Transform: Suppose known all cdfs:

$$(F_{x_{i_1}}, F_{x_{i_2}|x_{i_1}}, F_{x_{i_3}|x_{i_1}, x_{i_2}}, \dots, F_{x_{i_d}|\mathbf{x}_{\sim i_d}}), \forall i_k \in (1, 2, \dots, d).$$

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$$\begin{cases} x_{i_1} &= F_{x_{i_1}}^{-1}(u_{i_1}) \\ x_{i_2} &= F_{x_{i_2}}^{-1}(u_{i_2}|u_{i_1}) \\ \vdots &\vdots \\ x_{i_d} &= F_{x_{i_d}|\mathbf{x}_{\sim i_d}}^{-1}(u_{i_d}|\mathbf{u}_{\sim i_d}) \end{cases} \quad (2)$$

N.B.: **The Rosenblatt transformation is not unique** as there are $d!$ possible transformations.

Rosenblatt Transform

Example

We want a sample of $(x_1, x_2) \in (0, 1)^2$ uniformly distributed over the triangle $x_1 + x_2 \leq 1$. The problem being symmetric, we have $F_{x_1} = F_{x_2}$ and $F_{x_1|x_2} = F_{x_2|x_1}$.

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It can be shown that the marginal cdf is:

$$F_{x_i}(x_i) = 1 - (1 - x_i)^2 = u_i$$

and the conditional cdf is: $F_{x_j|x_i}(x_i, x_j) = \frac{x_j}{1-x_i} = u_j$

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The Rosenblatt transformation in this case is the following,

$$\begin{cases} x_i &= 1 - \sqrt{1 - u_i} \\ x_j &= u_j \sqrt{1 - u_i} \end{cases} \quad (i, j) = (1, 2) \text{ or } (i, j) = (2, 1)$$

Conclusion

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Independent case: If $\mathbf{x}$ is derived from $\mathbf{u} \sim \mathcal{U}(0, 1)^d$ by the integral transformation,

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Then,

$$y = \mathcal{M}(x_1, \dots, x_d) = \mathcal{M}(F_{x_1}^{-1}(u_1), \dots, F_{x_d}^{-1}(u_d)) = f(\mathbf{u})$$

The sensitivity of y to x_i is the same as its sensitivity to u_i .

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$$\begin{cases} x_{i_1} &= F_{x_{i_1}}^{-1}(u_{i_1}) \\ x_{i_2} &= F_{x_{i_2}}^{-1}(u_{i_2} | u_{i_1}) \\ \vdots &\vdots \\ x_{i_d} &= F_{x_{i_d} | \mathbf{x}_{\sim i_d}}^{-1}(u_{i_d} | \mathbf{u}_{\sim i_d}) \end{cases}$$

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The interpretation of the sensitivity of y to u_i as its sensitivity to x_i is more tricky.